

Amit Gohel

amitgohel2002@gmail.com | +91 70166 79944 | amitgohel.dev

linkedin.com/in/amitgohel-ai | github.com/amit-gohel

Professional Summary

AI/ML Engineer with 2+ years shipping production Generative AI (GenAI) systems at scale. Built RAG chatbots at 95% MRR handling 50,000+ queries/month, multi-model LLM pipelines across GPT-4o, Gemini, and Qwen, and AI-powered content generation systems. Deep expertise in Retrieval-Augmented Generation, prompt engineering, vector databases, and cloud deployment on AWS and Azure. Microsoft Azure AI Engineer Associate certified.

Technical Skills

- **Programming Languages:** Python, C++, SQL
- **Techniques:** Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Natural Language Processing (NLP), Text Summarization, Semantic Search, CNN, LSTM, BiLSTM
- **AI/ML Frameworks:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, LangChain, LangGraph, Generative AI models including GPT-4o, Gemini 2.5 Pro, Llama, Qwen (LLMs and large vision/multimedia models)
- **Vector Databases and Retrieval:** Milvus, FAISS, Qdrant, OpenAI Embeddings, BGE-M3, Cohere Embed/Rerank, SPLADE
- **Infrastructure and Cloud:** FastAPI, VLLM, AWS (EC2, Lambda, S3, SAM), Microsoft Azure, GCP, Docker, Git, PostgreSQL, Pydantic
- **Tools and APIs:** OpenAI API, Google Gemini API, Anthropic API, Together AI, ElevenLabs, Selenium, FFmpeg

Work Experience

AI Engineer, Vivansh Infotech *Ahmedabad, India* Aug 2025 – Present

- Upgraded the **permission-aware RAG chatbot** into a fully **agentic conversational assistant using LangChain and LangGraph** for a multi-tenant EHS platform serving **500+ concurrent users** and **1,000+ queries/day** – integrated with **Milvus**, **PostgreSQL**, and **10+ platform APIs**, enabling the chatbot to autonomously retrieve live data, trigger workflows, and resolve multi-turn queries end-to-end
- Enforced **250+ granular permissions per query** in real time via 2-stage retrieval: dedicated Milvus permission index cross-referenced against PostgreSQL, with **GPT-4o** handling access control, **prompt engineering for query rewriting**, and conversation resolution across 10,000+ content chunks; **80% MRR**
- Built **AI-powered form generation pipeline** converting PDFs to digital replicas in ~20 seconds (**200+ forms in month 1**); designed Pydantic-validated intermediate JSON schema enabling reliable LLM-to-platform conversion; implemented **multi-model Generative AI pipeline (GPT-4o, Qwen3-Coder-480B, GPT-4o-mini)** with each model selected via empirical benchmarking
- Created an **AI video course generation pipeline** reducing production time from ~1 week to ~10 minutes (99% reduction) by orchestrating RAG-based semantic search and video selection (Milvus + Cohere), **Gemini 2.5 Pro** for visual-audio alignment, ElevenLabs narration, and serverless FFmpeg rendering on **AWS Lambda**

AI/ML Developer, Silver Touch Technologies Ltd *Ahmedabad, India* Jun 2024 – Jul 2025

- Rebuilt production **Generative AI RAG chatbot** infrastructure serving **50,000+ queries/month** across 10+ chatbot instances – migrated from in-code Llama-2-7b to VLLM-served API and replaced FAISS with Milvus, improving MRR by **111% (45% to 95%)**
- Led vector database evaluation (Milvus vs. Qdrant vs. FAISS) on production-scale data; tested 6+ embedding models (BGE-M3, BGE-small, BGE-base, BGE-large, Stella, SPLADE) and built custom semantic search and retrieval logic with BGE-M3 + reranker
- Reduced chatbot response latency by **50% (16s to 8s)** through NLP-based query preprocessing with intent identification and retrieval pipeline optimization
- Built **on-premise OCR and document processing pipeline** for sensitive government documents (200+ page PDFs) – combined DocTR + DocLink OCR outputs via Llama-3.1-Nemotron-70b for hybrid merging, achieving ~90% entity extraction accuracy on Aadhaar, PAN, and GST numbers

Research Intern, Indian Institute of Technology (IIT) Delhi *New Delhi, India* Jan 2024 – May 2024

- Built end-to-end **air writing recognition system**: ESP32 + MPU6050 wearable classifying 62 alphanumeric characters

via CNN-BiLSTM deep learning model achieving 75% user-dependent accuracy

- Collected 12,400 labeled samples from 20 participants; systematically evaluated **1,000+ model configurations** across 5 architecture families (Dense, CNN, LSTM, BiLSTM, hybrids) using TensorFlow/Keras

Education

Bachelor of Engineering – Computer Engineering

Jun 2020 – Jun 2024

Vishwakarma Government Engineering College, Ahmedabad, India

Certifications and Achievements

- **Microsoft Azure AI Engineer Associate** – Certified in designing and implementing Azure AI solutions including Generative AI, NLP, and knowledge mining
- **Samsung Solve for Tomorrow** – Top 10 among 70,000+ participants nationwide; featured on CNN News18
- **Smart India Hackathon 2023** – Top 5 among 44,000+ teams nationwide (Hardware Edition, MIT Karnataka)
- **Competitive Programming**: 370+ problems solved in Python